DRUGS · ANTIBIOTICS: OTHER CLASSES

Antituberculous drugs

Tuberculosis reaches the ENT clinic as a neck lump far more often than as anything else, but also as laryngeal tuberculosis mimicking cancer, painless otorrhoea with multiple perforations, and sinonasal or oral ulceration. The treatment is a fixed combination given for at least 6 months under directly observed or closely supervised therapy by a tuberculosis team. The ENT surgeon's part is the diagnosis, the samples that make drug-susceptibility testing possible, and knowing the drugs' toxicities, including the ototoxicity of streptomycin and amikacin.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

ANTITUBERCULOUS DRUGS

	SUMMARY
First-line drugs	Rifampicin (R), isoniazid (H), pyrazinamide (Z), ethambutol (E)
How they work	Rifampicin blocks bacterial RNA polymerase; isoniazid blocks mycolic acid synthesis; pyrazinamide acts in acidic intracellular sites; ethambutol blocks arabinogalactan synthesis
Standard regimen	2 months of RHZE (intensive phase) then 4 months of RH (continuation phase): 6 months in total for lymph node, laryngeal and most extrapulmonary tuberculosis; 12 months for central nervous system disease
Adjuncts	Pyridoxine 10 mg daily with isoniazid to prevent neuropathy; steroids for tuberculous meningitis and pericarditis
Drug-resistant disease	Bedaquiline, linezolid, fluoroquinolones, delamanid, pretomanid and injectables (amikacin) under specialist care

	SUMMARY
Doses	By weight: rifampicin about 10 mg/kg, isoniazid 5 mg/kg, pyrazinamide 25 mg/kg, ethambutol 15 mg/kg once daily, given as fixed-dose combination tablets
Pregnancy	First-line drugs are used; streptomycin is contraindicated (fetal ototoxicity)

2 CHAPTER TWO How they work

Mechanism and handling.

2.1 Why combinations and long courses

Mycobacteria divide slowly and exist in different populations: actively dividing extracellular bacilli (killed by isoniazid), slowly metabolising intracellular bacilli in acidic phagosomes (pyrazinamide), and intermittently active bacilli (rifampicin). Combinations prevent the emergence of resistance from spontaneous mutations; six months are needed to sterilise the lesions and prevent relapse.

3 CHAPTER THREE Using them in ENT

What ENT surgeons do.

3.1 Diagnosis first

For a suspected **tuberculous lymphadenitis** (scrofula: a matted, painless, often cold neck mass with possible sinus formation), obtain tissue: ultrasound-guided core or FNA with samples for **mycobacterial culture and PCR** (GeneXpert, which also detects rifampicin resistance) as well as histology, or an excision biopsy. Do not incise and drain a tuberculous node: a chronic discharging sinus results. Send a chest radiograph, HIV test and contact history. Notify the case.

3.2 Paradoxical reactions

Nodes may enlarge or new ones appear during the first months of treatment (paradoxical upgrading reaction) in up to a fifth: this is not treatment failure. Aspirate fluctuant nodes; consider steroids for severe reactions.

3.3 Non-tuberculous mycobacteria in children

Atypical mycobacterial lymphadenitis (violaceous skin over a node in a well child aged 1 to 5) is not tuberculosis: excision is curative and antituberculous drugs are ineffective; clarithromycin with rifampicin is used when excision is unsafe.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

- **Hepatotoxicity** with rifampicin, isoniazid and pyrazinamide: baseline liver function and review if symptoms or abnormal results.
- **Rifampicin**: orange-red urine, tears and secretions (stains contact lenses); flu-like syndrome; potent enzyme induction.
- **Isoniazid**: peripheral neuropathy (prevent with pyridoxine), hepatitis, drug-induced lupus.
- **Pyrazinamide**: hyperuricaemia, gout, arthralgia, hepatotoxicity.
- **Ethambutol**: **optic neuritis** with loss of colour vision and acuity; baseline visual acuity and colour testing, and stop at once if vision changes.
- **Streptomycin and amikacin**: vestibular and cochlear ototoxicity, nephrotoxicity; audiometry during treatment.

4.2 Interactions

Rifampicin induces CYP enzymes and lowers the levels of **oral contraceptives**, warfarin, DOACs, steroids (double the steroid dose), antiretrovirals, ciclosporin,

azoles, methadone and many others.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- The standard **RHZE regimen** and its duration.
- How to sample a suspected tuberculous node (culture, PCR, not incision and drainage).
- The key toxicities: liver, **ethambutol and vision**, isoniazid and neuropathy, **streptomycin and the ear**.
- Rifampicin interactions, especially contraception.
- Atypical mycobacteria in children: excision, not antituberculous drugs.

5.2 Common pitfalls

COMMON PITFALLS

- Sending a node only for histology in formalin, with nothing for culture.
- Incising and draining a tuberculous node.
- Mistaking a paradoxical reaction for treatment failure.
- Forgetting pyridoxine or baseline vision testing.

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